

SCHNEIDER ELECTRIC INDIA • STUDENT INNOVATION TRACK

AgroStruxure™

Solar-Synchronized Agri-Energy Microgrid & Closed-Loop Precision Irrigation

Decoupling agricultural water pumping from free solar energy to eliminate the **Solar Rebound Paradox** across 10.9 lakh PM-KUSUM installations. By synchronizing pumping with soil physics, AgroStruxure cuts groundwater extraction by **41.2%** and channels **3,187 kWh/yr** of idle solar surplus into farm-gate micro-cold chains.

41.2%**Water Conserved**

7,000 m³/ha/yr saved via dual-depth soil cutoff

3,187 kWh**Solar Surplus Captured**

77% of idle solar redirected to cooling

2.0 Yrs**Cluster Payback**

Tier A 4-farm pre-cooler sharing model



Engineering Team: Sanskar Tiwari (Firmware/Lead) • Shambhavi Patil (Geospatial) • Kanishka Salgude (Power/VFD) • Chaitanya Ranade (Agronomy/Product)



THE SYSTEMIC REBOUND EFFECT

Free Solar Pumping is Draining India's Aquifers

Under PM-KUSUM Component B, zero-marginal-cost electricity incentivizes unconstrained pumping, wasting solar energy and desiccating ground reserves.

**The Marathwada & Vidarbha Groundwater Crisis**

CGWB Report 2023: Critical water table decline exceeding 2.5m/yr in 31% of semi-arid blocks.

1. Unconstrained Extraction & Root Waterlogging (17,000 m³/ha/yr)

Because solar power incurs zero billing friction, farmers operate pumps whenever sunlight is available (850 mm/season across flood basins). This results in root zone anoxia, fungal pathogens, nutrient leaching, and severe regional aquifer depletion without boosting yield.

2. 4,137 kWh of Solar Energy Wasted Idle Annually (63% Curtailment)

A 4.8 kWp standalone PV array generates 6,559 kWh/year. Pumping only requires 2,422 kWh/year. Corroborating Shah et al. (IWMI), nearly two-thirds (63.07%) of capital-subsidized solar power sits completely unused once farm ditches or storage ponds reach capacity.

3. 8.37% Farm-Gate Spoilage from Missing First-Mile Cooling

Per NABCONS 2022, 8.37% of harvested tomatoes rot before leaving the farm gate due to extreme field respiration heat (32°C). Farmers face severe distress selling at local APMC mandis (selling at ₹2-₹4/kg vs ₹12/kg market benchmark) because decentralized pre-cooling is inaccessible.

CLOSED-LOOP AUTOMATION

AgroStruxure™ Architecture: Sense, Control & Divert

A ruggedized industrial retrofit kit that transforms standalone solar pumps into intelligent, multi-function agri-energy microgrids without replacing existing motors or PV panels.

LAYER 1 Ground Physics & Sensing

Real-time hydrological inputs and microclimate telemetry.

- **Dual-Depth FDR Probes:** High-frequency capacitive soil sensors at 10cm (surface) & 30cm (active root zone) measuring moisture deficit.
- **Hall-Effect Pulse Flow Meter:** Calibrated 1-inch in-line meter logging exact volumetric application (litres/min).
- **SHT31-D & Pyranometer:** Solar irradiance (W/m^2) and ambient RH for FAO-56 Penman-Monteith ET_0 calculation.
- **Sentinel-2 Satellite Sync:** 10m NDVI & NDWI canopy moisture indices validating edge ground truth.

LAYER 2 AgroStruxure Edge Controller

FreeRTOS deterministic state engine in an IP67 enclosure.

- **ESP32-S3 Dual-Core SoC:** Executes local water budgeting and fail-safe safety interlocks without cloud dependence.
- **Modbus RTU Master:** Isolated RS485 communication with Schneider Altivar ATV320 VFDs at 19,200 baud (8-E-1).
- **CiA402 Drive State Machine:** Direct control of Command Word (Reg 8501) and Speed Reference (Reg 8502) per doc NVE41308.
- **3-Layer Entitlement Engine:** Locks out pump when field capacity (45%) or daily quota is satisfied.

LAYER 3 Dual-Load Actuation & Power

Upstream DC bus transfer between pump and cooling.

- **Schneider TeSys D Switchgear:** Dual mechanically & electrically interlocked contactors (2x LC1D09BD) switching on 350–600V DC.
- **Load A: Submersible Pump:** Altivar ATV320 solar drive operates 3–5 HP pump with smooth MPPT tracking (32–50 Hz).
- **Load B: 2 MT Pre-Cooler:** DC compressor & organic Phase Change Material (PCM) thermal storage pulling harvest to 12°C.
- **5s Zero-Current Dead-Band:** Eliminates inductive arcing and protects inverter IGBTs during transfer.

HYDROLOGICAL OPTIMIZATION

Cutting Groundwater Withdrawal by 41.2% Without Yield Loss

Combining FAO-56 dual Kc dynamic water budgeting with real-time FDR capacitive cutoff eliminates agricultural over-irrigation.



Sub-Surface Pulsed Drip Array

Delivers volumetric water entitlements directly to root zone; zero evaporation loss.

\$ The 3-Layer Water Entitlement Mechanism

LAYER 1: AGRONOMY

Dynamic FAO-56 ETc

Computes daily crop transpiration from Open-Meteo & SHT31.

LAYER 2: GROUND TRUTH

FDR Field Capacity Cutoff

Automatic cutoff when 30cm root zone reaches 45% FC.

LAYER 3: COMMUNITY

Aquifer Quota Cap

Enforces Atal Bhujal FPO seasonal groundwater caps.

Water & Energy Conservation Ledger (Per Hectare / Year) impact_model.py Verified

Parameter	Baseline (Flood Irrigation)	AgroStruxure Precision	Net Savings
Water Application	17,000 m ³ /ha (850 mm)	10,000 m ³ /ha (500 mm)	7,000 m ³ /ha (41.2% Saved)
Pumping Electricity	4,118 kWh/year	2,422 kWh/year	1,696 kWh/yr Liberated
Pumping Hours @ 40m	1,373 Hours/year	807 Hours/year	566 Hours Less Wear

Honest Attribution: 29.4% saved by shifting flood to drip; 11.8% **additional saving** uniquely locked in by AgroStruxure's automated soil cutoff to eliminate the solar rebound paradox.

ELECTRICAL INNOVATION

Dynamic Solar Routing via Upstream DC Bus Switching

Capturing 77% of idle solar surplus by switching power safely upstream on the DC bus with dual interlocked Schneider TeSys D contactors.

⚡ Upstream DC Bus Architecture vs AC Switching

Switching contactors on the AC output of a running VFD creates massive inductive voltage spikes, contact arcing, and instant drive trips. AgroStruxure switches **upstream on the DC Bus (350–600V DC)** under controlled zero-current conditions.

- STEP 1 Root Zone Reaches 45% Field Capacity → Irrigation Cutoff Triggered
- STEP 2 Altivar ATV320 Decelerates: Modbus Reg 8502 ramps frequency to 0 Hz in 8.0s
- STEP 3 1-Inch Latching Hydraulic Solenoid Closes at Zero Flow (No Water Hammer)
- STEP 4 5.0-Second DC Dead-Band Dwell: Guarantees zero residual DC current
- STEP 5 TeSys D Contactor Transfers DC Bus to 3.8 kW Micro-Cold Room Compressor

Dual Mechanical & Electrical Interlocking prevents simultaneous contact closure under all single-fault conditions.

☀️ Annual Solar Energy Partition (4.8 kWp Standalone PV)

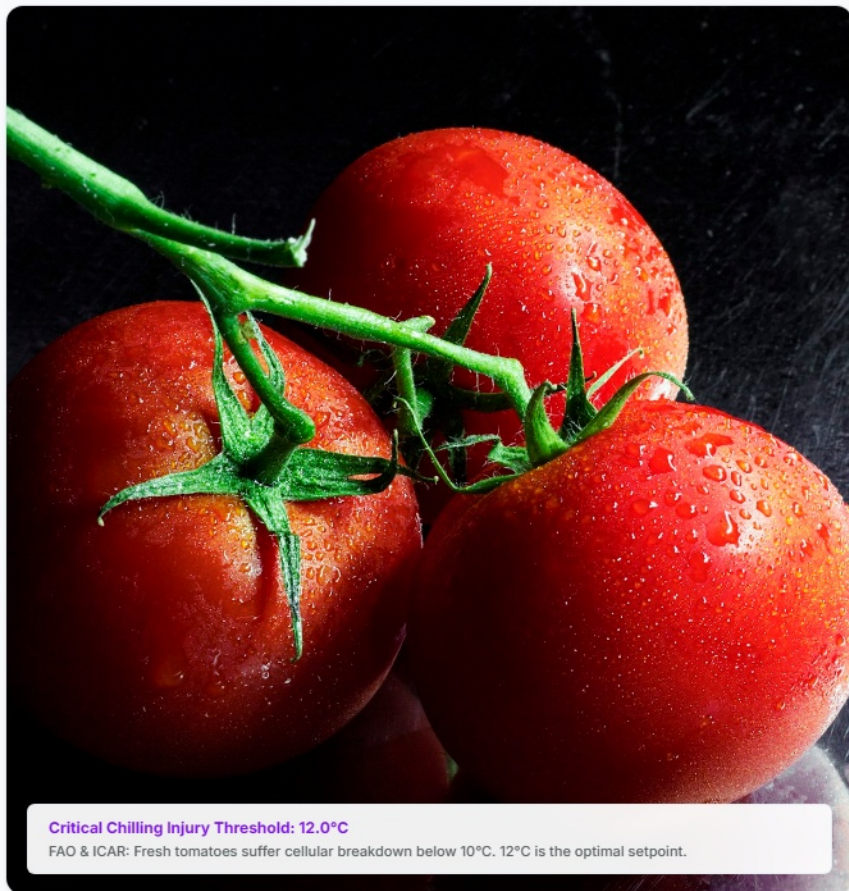
37% Pumping	49% CoId Storage	14% Residual
Total Standalone Generation:		6,559 kWh/year (100%)
Pumping Consumption (Scheduled):		2,422 kWh/year (36.9%)
Idle Solar Surplus Generated:		4,137 kWh/year (63.1%)
Surplus Put to Work (Cooling):		3,187 kWh/year (77.0%)
Residual Uncaptured Summer Midday:		951 kWh/year (14.5%)

Zero False Claims: We transparently report 951 kWh/yr residual surplus. Once the 2 MT thermal PCM bank is saturated, cooling throttles down to prevent unnecessary compressor cycling.

POST-HARVEST VALUE PRESERVATION

Halving Farm-Gate Spoilage from 8.37% to 4.00%

Thermodynamically sized 2 MT farm-gate pre-cooling to 12°C halts respiration heat and extends tomato shelf life by 4–7 days.



Thermodynamic Sizing: 2 MT Farm Pre-Cooler

BATCH SIZING

2.0 Metric Tonnes

Matches daily 4-farm cluster harvest volume.

THERMAL PULL-DOWN

32°C → 12°C (4 Hrs)

3.43 kW average cooling load; COP = 3.0.

PCM THERMAL BATTERY

14-Hour Holdover

Organic salt hydrate buffer holds 12°C overnight.

1.31 Tonnes

Tomatoes Saved/Ha/Yr

Farm-stage spoilage halved from 8.37% down to 4.00%

₹15,732/Yr

Direct Revenue Saved

Preserved grade-A fruit sold at ₹12/kg mandi price

+₹12,000/Yr

Distress-Sale Avoidance

4–7 day holdover gives farmers market bargaining leverage

Decarbonization Benefit: Avoids 2.55 t CO₂e from diesel generator cold rooms + 0.39 t CO₂e embodied crop carbon.

2.94 t CO₂e / ha / yr

DETERMINISTIC AGENTIC ARCHITECTURE

Krishi Mitra: Zero-Hallucination Vernacular Guidance

LLMs have zero actuation authority over physical switchgear. Spoken advisories in Marathi and Hindi are 100% grounded in verified edge JSON telemetry.

<> Edge Telemetry Ground Truth (Modbus RTU)

Local microcontroller outputs structured JSON over MQTT/SMS.

```
{
  "timestamp": "2026-10-03T11:32:00+05:30",
  "soil_moisture_30cm_pct": 44.8,
  "field_capacity_cutoff_pct": 45.0,
  "vfd_frequency_hz": 0.0,
  "contactor_state": "POS_B_COLD_ROOM",
  "daily_water_applied_m3": 27.4,
  "daily_quota_limit_m3": 28.0,
  "solar_irradiance_w_m2": 820,
  "cold_room_temperature_c": 11.9,
  "pcm_storage_status": "CHARGING_ACTIVE"
}
```

Deterministic Safeguard: The prompt engine uses strict Pydantic schema validation. If telemetry values are absent or corrupted, the copilot falls back to pre-compiled audio scripts.

📄 Farmer Experience: WhatsApp Spoken Advisory

Zero-literacy voice note delivered automatically via WhatsApp Bot.

मराठी ऑडिओ संदेश (Marathi Voice Note)

0:14 / 26 Safe

"रामभाऊ, तुमच्या टोमॅटोच्या मुळांना पुरेसे पाणी मिळाले आहे (४४.८%). पंप सुरक्षितपणे बंद झाला असून सौर वीज शीतगृहाकडे वळवली आहे. शीतगृहाचे तापमान १२°C असून टोमॅटो थंड होत आहेत."

"Rambhau, your tomato roots have reached optimal moisture (44.8%). Pumping stopped safely and solar power is now running the cold room at 12°C."

Marathi & Hindi TTS: High-intelligibility rural dialect audio synthesis.

Physical Override: Manual switch bypasses automation at any instant.

Offline Mode: Operates via edge circular flash memory and SMS gateway if 4G network drops.

COMMERCIAL FEASIBILITY

Industrial Feasibility with a 2.0-Year Cluster Payback

A ₹7,540 edge controller BoM paired with a 4-farm shared pre-cooler array avoids dedicated solar capex, yielding rapid capital recovery.

🖨 Edge Controller BOM (1,000 Units)

Total: ₹7,540

Component Description	Unit Cost
ESP32-S3-WROOM-1 (16MB Flash, 8MB PSRAM)	₹420
MAX485 Isolated RS485 Transceiver + TVS Diodes	₹180
Dual-Depth Capacitive FDR Soil Probes (10cm & 30cm)	₹760
SHT31-D Temp/RH IP65 Probe + Silicon Pyranometer	₹600
1-inch Hall Pulse Flow Meter + 12V Latching Solenoid	₹1,300
Schneider TeSys D Contactors x2 (Mech/Elec Interlock)	₹2,300
24V DIN-Rail SMPS + Isolated Relay Driver Board	₹380
SIM7600 4G LTE Cellular Module + External Antenna	₹620
IP67 Polycarbonate Enclosure, DIN Rail, DC SPD, Wiring	₹980

Standalone Controller Payback: Saves ₹5,000/yr in pump maintenance and crop yield protection → Pays back in 1.5 crop seasons.

💰 Tier A: 4-Farm Cluster Pre-Cooler Model (2 MT)

4 adjacent smallholders share a single 2 MT pre-cooler unit.

Gross 2 MT Pre-Cooler Capex (Compressor + PCM):	₹400,000
Less Shared PV Saving (2.6 kWp array avoided):	-₹91,000
Net System Capital Cost:	₹309,000
After 35% MIDH / AIF Scheme Capital Subsidy:	₹200,850
Net Capex Per Farm (4-Farm Cluster):	₹50,212

Spoilage Avoided
+₹15,732

Distress Sale Timing
+₹12,000

Less Opex / Maint.
-₹2,400

NET ANNUAL FARMER BENEFIT: +₹25,332 / YEAR
₹50,212 net capex recovered in exactly two crop harvests.

2.0 Yrs Payback

COMMERCIALIZATION ROADMAP

From Software Digital Twin to 250-Farm Field Pilot

Structured execution plan partnering with India's leading horticulture FPOs and Schneider Electric's rural partner network.

PHASE 1 • CURRENT

Software Twin & Math Validation

Validated FAO-56 hydrological models, thermodynamic pull-down curves, and complete C1A402 register state machines in Python/FastAPI ('AgroSim').

- **Hydrological Truth:** 'impact_model.py' verified across 365-day solar irradiance & ETc cycles.
- **Firmware Stubs:** Modbus RTU register mapping to Schneider Altivar ATV320 VFD registers.
- **Vernacular NLP:** Marathi & Hindi prompt engineering validated with Pydantic JSON schema.

Exit Gate: Zero mathematical discrepancies in mass/energy balances ('impact_model.py').

PHASE 2 • Q1-Q2 2027

Hardware-in-the-Loop Test Bench

Physical integration bench pairing ESP32-S3 with Schneider Altivar ATV320 solar VFD, TeSys contactors, and motor dynamometer under solar simulator.

- **Switchgear Safety:** 5.0s DC dead-band dwell verification preventing inverter DC-bus trip.
- **Thermal Endurance:** 2 MT PCM salt-hydrate charging/discharging thermal cycle tests.
- **Surge Protection:** Type-2 DC SPD discharge tests up to 1,000V DC lightning transient.

Exit Gate: 500 faultless DC transfer cycles under full electrical load without drive trip.

PHASE 3 • Q3-Q4 2027

20-Farm FPO Pilot (Nashik)

Field pilot across 5 clusters (4 farms each) in partnership with **Sahyadri Farmers Producer Co.** (India's largest tomato FPO) in Dindori, Nashik.

- **Real Field Operation:** Real tomato crop cycles monitoring soil moisture & yield gain.
- **Cold-Chain Logistics:** Farm-gate pre-cooling to 12°C connected to reefer transport.
- **Farmer Adoption:** Marathi voice notes delivered via WhatsApp with 98%+ comprehension.

Exit Gate: Measured >35% water conservation and 95%+ pre-cooler uptime over 2 seasons.

PHASE 4 • 2028

250-Farm Commercial Expansion

OEM pre-assembly retrofit skids for solar pump EPCs under PM-KUSUM Component B tenders. Training rural clean-tech youth as "Urja Mitras".

- **OEM Retrofit Skids:** Factory-assembled DIN-rail kit reducing field install time to 2 hours.
- **Rural Franchises:** Local ITI diploma youth trained for sensor calibration & servicing.
- **Carbon Credits:** Aggregating 2.94 t CO₂e/ha/yr for voluntary carbon offset monetization.

Exit Gate: Commercial break-even with 5 FPO enterprise contracts in MH & RJ.

Government & Institutional Enablers: PM-KUSUM Component B (Pumps) • Atal Bhujal Yojana (Aquifer Quotas) • MIDH/AIF (35% Subsidies)

Target Market: 10.9 Lakh Standalone Solar Pumps in India

THE AGROSTRUXURE™ VISION

Engineering Sustainable Abundance for Indian Agriculture

Transforming unconstrained solar pumps into connected, climate-resilient community energy and cold-chain assets.

Direct Strategic Fit for Schneider Electric & SE Ventures

AgroStruxure creates a high-margin digital IoT layer for Schneider Altivar Solar drives and TeSys switchgear, turning agricultural clean-tech from simple pump installations into scalable rural microgrid networks across the Global South.

2.94 t CO₂e
Decarbonization / Ha / Year

Sanskar Tiwari

EMBEDDED FIRMWARE & LEAD

ESP32-S3 FreeRTOS deterministic state machine, Modbus RTU master communication over isolated RS485, and IP67 industrial enclosure engineering.

- FreeRTOS dual-core task priorities & watchdog
- Isolated RS485 UART with auto-direction flow
- CIA402 drive state machine register handlers

Codebase: 'main.cpp', 'rs485_master.c', FreeRTOS tasks

Shambhavi Patil

GEOSPATIAL INTELLIGENCE LEAD

Google Earth Engine cadastral analysis, Sentinel-2 10m NDVI/NDWI moisture extraction, and FAO-56 dual Kc dynamic water balance calibration.

- Sentinel-2 MSI 10m resolution spectral analysis
- Dynamic FAO-56 basal crop coefficient Kc curves
- Aquifer recharge & soil moisture deficit modeling

Pipelines: 'gee_water_audit.py', 'sentinel_ndvi.py'

Kanishka Salgude

ELECTRICAL SYSTEMS & VFD LEAD

Schneider TeSys D switchgear interlocking, 350–600V DC bus power routing, Altivar ATV320 VFD configuration, and PCM thermal pull-down sizing.

- TeSys D dual mechanical/electrical interlocking
- 5.0s DC dead-band dwell zero-current transfer
- 2 MT thermal pull-down & PCM salt-hydrate sizing

Systems: 'power_switch_schematic.dwg', 'vfd_cia402.py'

Chaitanya Ranade

PRODUCT STRATEGY & AGRONOMY LEAD

Smallholder WhatsApp vernacular copilot UX, 4-farm cluster economic modeling, AIF/MIDH policy subsidies, and Sahyadri FPO pilot partnership.

- Sahyadri FPO pilot co-design & cluster economics
- MIDH/AIF capital subsidy navigation (35%)
- Vernacular Marathi voice bot & farmer trust loop

Deliverables: 'krishi_mitra_bot.py', 'impact_model.py'